

Digital Quantum Simulation of Nonequilibrium Dynamics in the Schwinger Model under a Strong External Electric Field

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We use the (1+1)-dimensional Schwinger model to investigate the nonequilibrium dynamics of a finite lattice system under a constant external electric field. The lattice Hamiltonian is constructed under open boundary conditions. The vacuum state is prepared using the variational quantum eigensolver (VQE). Scans over the external field strength show the flip of the vacuum state at several field strengths. The critical field strengths agree with theoretical predictions. We further investigate the real-time evolution of the zero-field vacuum under an external electric field using a second-order Trotter-Suzuki decomposition. By comparison with exact diagonalization (ED), we verify that the quantum-simulation protocol reproduces the main features of field-induced boundary charge separation, decay of the vacuum-state fidelity, and quasiperiodic energy redistribution between the electric-field energy term and the fermionic sector. Our results indicate that combining VQE-based state preparation with digital real-time evolution provides a useful approach for studying nonequilibrium dynamics in strong-field lattice gauge theories.

I. INTRODUCTION

Vacuum instability in strong-field quantum electrodynamics is an important problem in nonperturbative quantum field theory. In a sufficiently strong electric field, the quantum vacuum can decay through tunneling and produce particle-antiparticle pairs, a process known as the Schwinger effect [1–5]. This process exhibits a characteristic nonperturbative exponential dependence and is intrinsically a real-time nonequilibrium phenomenon [3, 6–8]. It is therefore difficult to describe reliably using traditional perturbation theory. Numerical methods capable of directly tracking the real-time evolution of quantum states are essential for understanding vacuum structure, pair production, and energy transfer mechanisms in strong-field QED [32, 34, 57, 58].

Lattice gauge theory provides a first-principles framework for studying nonperturbative quantum field theory [9–13]. In Euclidean path-integral Monte Carlo simulations, efficient importance sampling is possible when the weight can be interpreted as a non-negative probability distribution. However, when the weight becomes complex, non-positive, or highly oscillatory because of a chemical potential, a topological term, a fermion determinant, or a real-time phase factor, severe cancellations occur among different field configurations, and the statistical error grows rapidly with the system volume or the evolution time. This

difficulty is usually referred to as the sign problem [14–16]. The Hamiltonian formalism is not by itself a source of quantum speedup, but it does not rely on Euclidean probability-weight sampling and can therefore bypass this class of sign problems associated with non-positive sampling weights. The corresponding difficulty is then shifted to the direct treatment of an exponentially large Hilbert space. Quantum computation and quantum simulation provide natural platforms for real-time lattice field theory, since quantum many-body states and unitary time evolution can be directly represented with qubits [17–20].

The Schwinger model, namely quantum electrodynamics in 1+1 dimensions, is a standard low-dimensional model for studying strong-field nonequilibrium dynamics [21, 22]. Although it is much simpler than QED in 3+1 dimensions, it retains several essential features, including local U(1) gauge invariance, charge screening, confinement, chiral condensation, and vacuum instability [23–34]. In addition, gauge fields in one spatial dimension have no independent degrees of freedom. Under open boundary conditions, Gauss’s law allows the electric field on each link to be expressed as a function of the matter charge distribution and the boundary electric field, thereby reducing the model to a purely fermionic Hamiltonian or, after a Jordan-Wigner transformation, a qubit Hamiltonian [25, 26, 35–37]. This feature makes the Schwinger model a useful benchmark platform for digital quantum simulation of lattice theories [38–46].

In recent years, the Schwinger model has become an important test bed for quantum simulations in

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high-energy physics. The real-time dynamics, pair production, thermalization, phase structure, and finite-temperature properties of the Schwinger model and related lattice models have been studied using platforms such as trapped ions, superconducting qubits, tensor networks and cold atoms [41, 42, 47–61]. In particular, quenches in which an external electric field is suddenly switched on have been used to investigate strong-field-induced particle-antiparticle production and vacuum response [32, 57, 58]. In related studies, the zero-field ground state is often prepared using VQE, evolved under the Hamiltonian with an external field, and compared with results from exact diagonalization or tensor-network calculations [32, 58, 62–65].

In this work, we study the lattice Schwinger model with a constant external electric field and focus on the finite-size nonequilibrium dynamics after an electric-field quench. We first **construct the lattice Hamiltonian under open boundary conditions and map it to a Pauli-operator form using the Jordan-Wigner transformation.** We then **use VQE to prepare the zero-field vacuum state, which is taken as the initial state when the external field is switched on.** The **real-time evolution is implemented by a second-order Trotter-Suzuki decomposition.** In contrast to approaches based on the McLachlan variational principle and fixed-depth variational real-time evolution, the **present work uses explicit digital Trotter evolution and examines the total charge, total energy, spatial-point charge distribution, vacuum-state fidelity, and electric-field energy** [57, 58, 68–72]. By comparing these quantities with ED results, we assess the **reliability of the protocol for describing strong-field dynamics in finite lattice systems.** The numerical implementation is based on the Qiskit quantum computing framework [73, 74]. We also perform **scans over the external field and analyze the chiral condensate, the low-energy levels, and the finite-size critical external field.**

The paper is organized as follows. Section II introduces the lattice Schwinger model with an external electric field, the Gauss-law reduction, and the qubit encoding. Section III describes the VQE state preparation and the second-order Trotter-Suzuki evolution. Section IV presents numerical results. Section V summarizes the work and discusses possible extensions, including larger system sizes, improved ansatzes, and implementation on noisy quantum hardware. Appendix A gives dynamical checks for the $N = 12$ lattice.

II. LATTICE SCHWINGER MODEL WITH AN EXTERNAL ELECTRIC FIELD

A. Continuum Model and Lattice Discretization

In natural units, the continuum Lagrangian density of the Schwinger model is [21, 22, 24]

$$\mathcal{L} = \bar{\psi} (i\gamma^\mu D_\mu - m) \psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad (1)$$

where $D_\mu = \partial_\mu + igA_\mu$ is the covariant derivative, ψ is the fermion field, m is the bare fermion mass, g is the gauge coupling, and A_μ is the $U(1)$ gauge field. In the temporal gauge $A_0 = 0$, the Hamiltonian can be written as

$$H = \int dx \left[-i\bar{\psi}\gamma^1 D_1\psi + m\bar{\psi}\psi + \frac{1}{2}E^2 \right]. \quad (2)$$

To simulate this theory with a finite number of qubits, the continuum field theory must be discretized on a finite lattice. In this work, we adopt the Kogut-Susskind staggered-fermion discretization, where the fermionic degrees of freedom reside on lattice sites and the gauge degrees of freedom reside on the links between neighboring sites [10, 25, 26].

Consider a system with N lattice sites and lattice spacing a , with the sites labeled by $n = 1, 2, \dots, N$. The discretized lattice Hamiltonian is

$$H = \frac{1}{2a} \sum_{n=1}^{N-1} (\phi_n^\dagger U_n \phi_{n+1} + H.c.) + m \sum_{n=1}^N (-1)^n \phi_n^\dagger \phi_n + \frac{g^2 a}{2} \sum_{n=1}^{N-1} L_n^2. \quad (3)$$

Here, ϕ_n is the staggered-fermion operator on the n -th lattice site, U_n is the gauge-link variable on the n -th link, and L_n is the corresponding lattice electric field. The first term describes hopping between neighboring lattice sites, the second term is the staggered mass term, and the third term is the electric-field energy. Unless otherwise specified, all numerical calculations in this work are performed in lattice units with $a = m = g = 1$.

B. Gauss's Law, Open Boundary Condition, and the External Electric Field

In $1+1$ dimensions, the gauge field has no independent propagating degrees of freedom, and the electric field on each link is fully determined by Gauss's law and the boundary conditions. The lattice Gauss law is [10, 25, 26]

$$L_n - L_{n-1} = \phi_n^\dagger \phi_n - \frac{1 - (-1)^n}{2}. \quad (4)$$

In this work, we impose open boundary conditions and set the electric field at the left boundary to a constant external field [24, 55, 58, 60]. From Gauss's law, one obtains

$$L_n = \varepsilon + \sum_{l=1}^n \left[\phi_l^\dagger \phi_l - \frac{1 - (-1)^l}{2} \right]. \quad (5)$$

Therefore, the external field ε is not introduced as an independent potential-energy term. Instead, it enters the Hamiltonian through the link and the corresponding electric-field energy [24, 55, 58]. It modifies the background electric field configuration and thereby affects both the ground-state structure and the real-time dynamics.

A further gauge transformation can absorb the link variable U_n in the hopping term [25, 26, 55, 58]. The Hamiltonian can then be written in a purely fermionic form as

$$H = \frac{1}{2a} \sum_{n=1}^{N-1} (\phi_n^\dagger \phi_{n+1} + H.c.) + m \sum_{n=1}^N (-1)^n \phi_n^\dagger \phi_n + \frac{g^2 a}{2} \sum_{n=1}^{N-1} \left[\varepsilon + \sum_{l=1}^n \left(\phi_l^\dagger \phi_l - \frac{1 - (-1)^l}{2} \right) \right]^2 \quad (6)$$

This expression shows that, after the gauge field is eliminated, the electric field energy becomes a long-range interaction induced by the charge distribution. The external field ε , together with the lattice charge distribution, determines the electric field on each link and is a key ingredient of the strong-field dynamics studied in this work.

C. Qubit Encoding

To implement the above model in digital quantum simulation, the fermionic operators must be mapped to Pauli operators. In this work, we use the Jordan-Wigner transformation [35–37],

$$\phi_n = \left(\prod_{l=1}^{n-1} i\sigma_l^z \right) \frac{\sigma_n^x - i\sigma_n^y}{2}. \quad (7)$$

After the Jordan-Wigner mapping, the lattice Schwinger model can be written as a Pauli Hamiltonian,

$$H = H_{\text{kin}} + H_m + H_E, \quad (8)$$

where the kinetic term is

$$H_{\text{kin}} = \frac{1}{4a} \sum_{n=1}^{N-1} (\sigma_n^x \sigma_{n+1}^x + \sigma_n^y \sigma_{n+1}^y), \quad (9)$$

the mass term is

$$H_m = \frac{m}{2} \sum_{n=1}^N (-1)^n \sigma_n^z, \quad (10)$$

and the electric-field energy term is

$$H_E = \frac{g^2 a}{2} \sum_{n=1}^{N-1} \left[\varepsilon + \frac{1}{2} \sum_{l=1}^n (\sigma_l^z + (-1)^l) \right]^2. \quad (11)$$

Since H_E contains the cumulative charge from the left boundary to the n -th lattice site, it generates long-range interactions in the Pauli representation. This nonlocal structure is a direct consequence of eliminating the gauge degrees of freedom in one spatial dimension, and it distinguishes the present model from ordinary nearest-neighbor spin chains [25, 26, 55, 58].

D. Observables

To characterize the nonequilibrium dynamics under a strong external electric field, we mainly focus on the following observables.

The total charge operator is defined as

$$Q_N = \frac{1}{2} \sum_{n=1}^N (\sigma_n^z + (-1)^n). \quad (12)$$

The total charge is a conserved quantity associated with the $U(1)$ gauge symmetry [51, 57, 58]. During real-time evolution, the conservation of the total charge can be used as a basic check of the correctness of the initial state, the Gauss-law constraint, and the time-evolution algorithm.

The local charge density on the n -th lattice site is defined as

$$q_n(t) = \frac{1}{2} (\langle \sigma_n^z(t) \rangle + (-1)^n). \quad (13)$$

In order to display the physical charge distribution more clearly, we also measure the local charge density on each spatial point, which is the summation of

charge densities on neighboring staggered lattice sites. For each spatial point i , the spatial-point charge is then defined as

$$Q_i(t) = q_{2i-1}(t) + q_{2i}(t). \quad (14)$$

This observable is used to characterize the spatial separation of positive and negative charges induced by the external electric field.

The corresponding total electric-field energy is

$$H_E(t) = \frac{g^2 a}{2} \sum_{l=1}^{N-1} \langle E_l^2(t) \rangle. \quad (15)$$

To quantify the stability of the initial vacuum state under the external field, we define the vacuum-state fidelity as [3, 32, 47, 58]

$$P_{\text{vac}}(t) = |\langle \psi_0 | \psi(t) \rangle|^2, \quad (16)$$

where ψ_0 is the zero-field ground state and $\psi(t)$ is the state evolved after the electric-field quench.

For the static scan over the external field, we also introduce the chiral condensate density [23, 24, 27–34, 59, 60],

$$\Gamma = \langle \bar{\psi} \psi \rangle = \frac{1}{2Na} \sum_{n=1}^N (-1)^n \langle \sigma_n^z \rangle, \quad (17)$$

which is used to identify field-induced ground-state reconstruction in finite-size systems.

III. NUMERICAL METHODS AND QUANTUM-SIMULATION PROTOCOL

A. Exact Diagonalization and Benchmark Calculations

For a given system size N , lattice spacing a , fermion mass m , gauge coupling g , and external-field strength ε , we construct the sparse-matrix Hamiltonian of the lattice Schwinger model and use exact diagonalization (ED) to obtain the ground state, the low-energy spectrum, and benchmark real-time dynamics. These results are used to assess the accuracy of the quantum algorithms.

B. VQE Ground-State Preparation and VQD Excited-State Estimation

The initial state for the real-time evolution is taken as the ground state of the zero-field Hamiltonian

$H_{\varepsilon=0}$. In this work, we use the variational quantum eigensolver (VQE) to prepare this initial vacuum state [62–65]. For a trial state $\psi(\theta)$ generated by a parameterized quantum circuit, VQE approximates the ground-state energy and wave function by minimizing the energy expectation value

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle. \quad (18)$$

We employ the hardware-efficient RealAmplitudes parameterized circuit and use SLSQP as the classical optimizer [64, 73, 74]. To reduce the influence of local minima, multiple random initial-parameter restarts are used during the optimization [67]. The accuracy of the VQE-prepared initial state is evaluated by comparing the ground-state energy and wave-function fidelity with the ED results. In addition to the zero-field initial-state preparation, VQE is also used in the static external-field scan to estimate the ground-state energy $E_0(\varepsilon)$ and the corresponding chiral condensate at different external-field strengths.

To further compare the low-energy structure, the first excited-state energy is estimated using the variational quantum deflation (VQD) method with an orthogonality penalty [66]. After obtaining the approximate VQE ground state ψ_0^{VQE} , the variational objective function for the first excited state is taken as

$$C_1(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle + \beta \left| \langle \psi_0^{VQE} | \psi(\theta) \rangle \right|^2, \quad (19)$$

where β is the orthogonality-penalty coefficient. The second term suppresses the overlap between the candidate state and the VQE ground state, thereby driving the optimization toward the first-excited-state subspace. After the optimization, the first excited-state energy is evaluated as

$$E_1^{VQD} = \langle \psi_1^{VQD} | H | \psi_1^{VQD} \rangle, \quad (20)$$

and its accuracy is assessed by comparison with the ED result E_1^{ED} . In the external-field scan, the optimized parameters from the previous field point are used as the initial parameters for the next field point to improve optimization stability. Thus, the quantum-algorithm results in the static external-field scan consist of two parts: VQE provides the ground-state energy and the chiral condensate, while VQD provides the first excited-state energy.

C. Second-Order Trotter-Suzuki Real-Time Evolution

After obtaining the zero-field ground state, we study the real-time dynamics after the external elec-

tric field is suddenly switched on. The initial state is taken as the ground state at $\varepsilon = 0$. After the quench, the system evolves under the Hamiltonian H_ε ,

$$|\psi(t)\rangle = e^{-iH_\varepsilon t} |\psi_0\rangle. \quad (21)$$

Since the Hamiltonian is decomposed into several noncommuting terms, the full time-evolution operator cannot be implemented directly [68–72]. If the total Hamiltonian is written as a sum of L terms, $H = \sum_{j=1}^L H_j$, and the total evolution time T is divided into r time steps, the second-order Trotter-Suzuki decomposition is given by

$$e^{-iHT} \approx \left[\prod_{j=1}^L e^{-iH_j \Delta t/2} \prod_{j=L}^1 e^{-iH_j \Delta t/2} \right]^r, \quad (22)$$

where $\Delta t = T/r$. The local error of each time step is $O(\Delta t^3)$, and the global error is second order in Δt for a fixed total evolution time. Compared with the first-order decomposition, the symmetric second-order formula provides higher accuracy for the same time step, at the cost of an increased circuit depth [68–72].

In the numerical implementation, the full Pauli Hamiltonian is passed to a second-order Suzuki-Trotter synthesizer, and Suzuki-Trotter (order=2, reps=1) is used to generate the evolution circuit for a single time step. The main time step is chosen as $\Delta t = 0.1$, and the total evolution time is taken in the range $T/a \in [0, 12]$. The representative external-field strengths in the dynamical calculations are selected according to the observable under consideration. In particular, the spatial-point charge distribution and electric-field energy response are mainly studied for $\varepsilon = 0.5, 1.0, 1.5, 2.0$, while the analysis of the vacuum-state fidelity and the effective decay rate is further extended to stronger external fields. The numerical implementation is based on Qiskit and the standard scientific Python packages [73–77]. The current simulation is performed on classical computers without considering quantum errors.

IV. NUMERICAL RESULTS

The main results are summarized as follows. For the system with $N = 8$ and $a = m = g = 1$, VQE prepares the ground state with high accuracy. We scan over the external field strength and calculate the chiral condensate and energy gap. These features are consistent with the flip of the finite-volume ground state. We then investigate the electric-field quench of the zero-field ground state. The second-order Trotter-Suzuki evolution preserves the total charge and total

energy and shows good agreement with the ED results. The external electric field drives oscillations of the spatial point charges near the open boundaries; the vacuum-state fidelity exhibits an effective exponential decay within the early-time window, and the electric field energy shows quasiperiodic behavior, reflecting finite-size energy redistribution between the electric-field energy term and the fermionic sector.

A. Ground-State Preparation and Study of the Critical Field Strength

We first examine the VQE preparation of the zero-field ground state for $N = 8$, $a = m = g = 1$, and $\varepsilon = 0$. The results are shown in Table I. The fidelity is defined as $F = |\langle \psi_0^{\text{ED}} | \psi_0^{\text{VQE}} \rangle|^2$. These results indicate that the chosen parameterized circuit can approximate the zero-field ground state with high accuracy at the present system size, providing a reliable initial state for the subsequent electric-field-quench dynamics.

We then turn on the external field, prepare the ground and first excited states. We calculate the chiral condensate $\langle \bar{\psi} \psi \rangle$ of the ground state, the ground state energy E_0 , and the first excited state energy E_1 , as shown in Figure 1.

The chiral condensate closely resembles a step function, and the jumps occur when the energy gap between the ground state and the first excited state vanishes. This result indicates that the external electric field can modify the vacuum structure of the finite system, driving the ground state from one to another. This phenomenon corresponds to a first-order phase transition at zero temperature, as observed in various studies of 1+1D models [24, 28, 59]. The VQE results are in overall agreement with the ED results.

It should be emphasized that the external-field scan discussed here is performed in finite-size systems. While the phase structure may remain qualitatively the same in the infinite-volume limit, the quantitative critical field strength is affected by finite-size effects. To further examine the finite-size dependence of this low-energy spectral reconstruction signal, we calculate the E_0 and E_1 for different lattice sizes N . Figure 2 shows the first critical field strength ε_c as a function of $1/N$. One can see that ε_c has a clear finite-size dependence on $1/N$, consistent with theoretical predictions. Doing a linear extrapolation, we get the critical field strength at infinite volume to be $\varepsilon_c(\infty) \approx 0.469$. This is smaller than the theoretical value $\frac{1}{2}$. We believe that this is due to the open boundary condition. It can be shown that the boundary term contributes negatively to the energy gap, thereby reducing the critical field strength.

TABLE I. Comparison between VQE and ED for ground state preparation

System parameters	ED energy	VQE energy	Relative error	Fidelity
$N = 8, a = m = g = 1, \varepsilon = 0$	-4.63805774	-4.63766032	8.569×10^{-5}	99.9931%

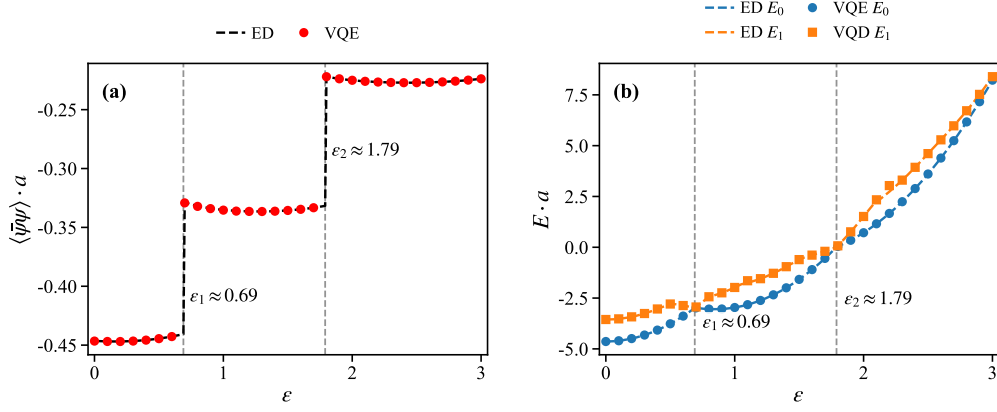


FIG. 1. Static observables as functions of the external-field strength ε . The left figure shows the chiral condensate of the ground state; the right figure shows the energy of the ground state and the first excited state.

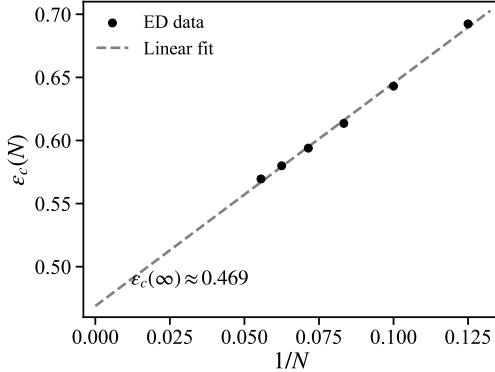


FIG. 2. Critical field strength ε_c as a function of $1/N$

B. Dynamical Evolution under an External Field

We proceed to the main part of this work. We prepare the ground state in the absence of an external field, then turn the field on at $t = 0$ and study the time evolution of the state.

First, we test the conservation laws after the electric-field quench. For different external-field strengths, the total charge $Q_N(t)$ remains zero throughout the time evolution. This indicates that the second-order Trotter-Suzuki evolution preserves the $U(1)$ charge as is expected.

We also calculate the total energy shift, $\Delta E(t) = E(t) - E(0)$. $\Delta E(t)$ remains very close to zero throughout the simulated time interval. This shows

that, with the chosen time step, the time evolution is numerically stable. The results are shown in Figure 3.

Although the total charge is conserved, the spatial charge distribution $Q_i(t)$ undergoes a strong rearrangement, as displayed in Figure 4. After the field quench, a charge-anticharge pair is periodically produced and annihilated on the boundary of the system. The period and amplitude of the oscillation increase with the external field strength. As a related quantity, we also calculate the time evolution of total electric field energy $H_E = \frac{g^2 a}{2} \sum_{l=1}^{N-1} \langle E_l^2(t) \rangle$ under different ε , as shown in Figure 5. When the charge pair is produced, energy is transferred from the electric field to the fermions, and when it is annihilated, the energy returns to the field. However, $H_E(t)$ does not show as simple a periodic pattern as $Q_i(t)$. This is related to the fact that at each spatial point, the fermion field has two components. The evolution of each component (included in the Appendix) is complicated as H_E , but they add up to a simple periodic mode as $Q_i(t)$. This indicates that the actual particle-antiparticle pair production is much more complicated than it may look like, with contributions from higher-energy modes.

It is also worth noting that the dependence of the period and amplitude on ε appears to be smooth, unlike that of the chiral condensate. This is not surprising, because we can see from Figure 1 that the eigen energies' dependence on ε is actually smooth.

To characterize the system's rate of deviation from the initial state, we calculate the vacuum-state fidelity $P_{\text{vac}}(t) = |\langle \psi_0 | \psi(t) \rangle|^2$. This quantity describes

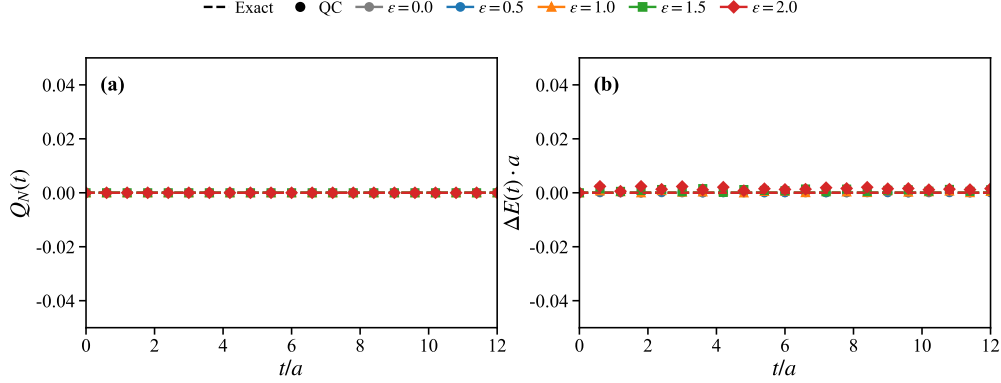


FIG. 3. Time evolution of (left) the total charge $Q_N(t)$ and (right) the total energy shift $\Delta E(t) = E(t) - E(0)$. In both panels, dashed lines denote ED results, while dots denote results obtained from the quantum algorithm.

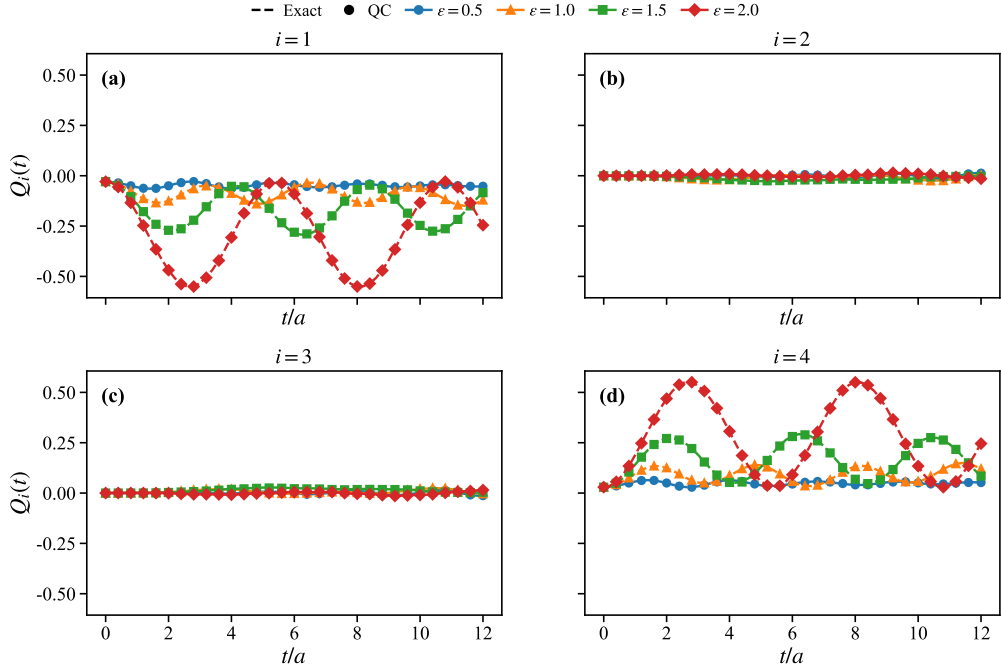


FIG. 4. Time evolution of the spatial charge distribution

the probability that the system remains in the initial vacuum state under the driving of the strong external electric field, and is used to characterize the field-induced deviation from the vacuum state.

Figure 6 shows that, for weak external fields, the vacuum-state fidelity remains close to unity. As the external field is increased, the system deviates from the zero-field vacuum more rapidly, indicating that a stronger external field enhances the nonequilibrium response of the system. Since the present study concerns a finite closed system, $P_{\text{vac}}(t)$ does not exhibit strictly monotonic decay, but instead shows finite-size revivals and quasiperiodic oscillations similar to

H_E . Again, the quantum algorithm results agree well with the ED results, demonstrating that the evolution protocol captures the vacuum response of the finite-lattice system.

To further quantify the initial decrease of the vacuum-state fidelity under different external fields, we perform an effective exponential fit to $P_{\text{vac}}(t; \epsilon)$ in the early-time window $0 \leq t/a \leq 1.0$: $P_{\text{vac}}(t; \epsilon) \simeq A(\epsilon) \exp[-\gamma_{\text{eff}}(\epsilon)t]$. Equivalently, we fit $-\ln P_{\text{vac}}(t; \epsilon) = \gamma_{\text{eff}}(\epsilon)t + c(\epsilon)$, and define the slope as the effective decay rate γ_{eff} . Within this early-time window, the numerical results show that $-\ln P_{\text{vac}}(t; \epsilon)$ is approximately linear in time, indi-

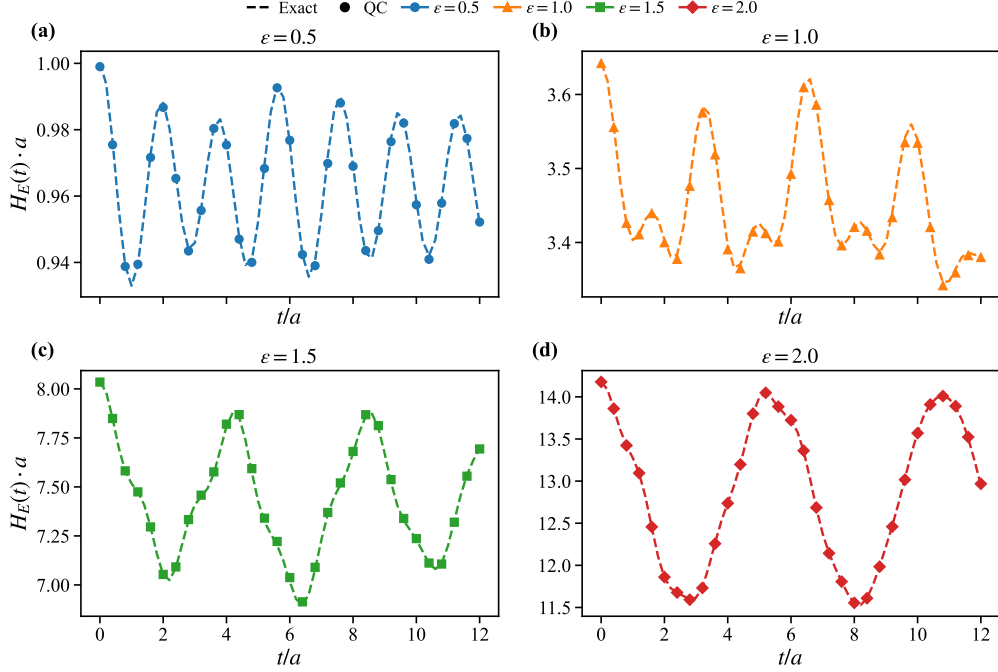


FIG. 5. Time evolution of the total electric field energy $H_E(t)$ under different external-field strengths

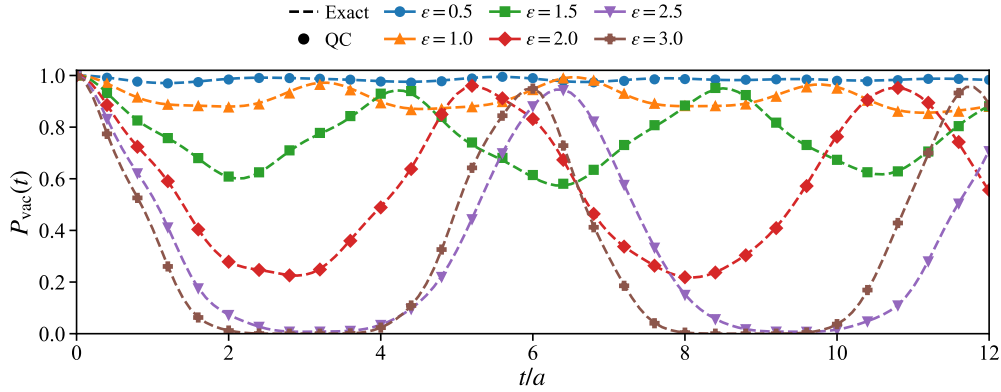


FIG. 6. Time evolution of the vacuum-state fidelity under different external field strengths

cating that the initial decrease of the vacuum-state fidelity can be characterized by an exponential form.

Figure 7 shows the dependence of γ_{eff} on the external-field strength ε extracted from the above fitting procedure. The effective decay rate increases monotonically with ε , indicating that a stronger external field drives the system away from the initial vacuum state more rapidly.

As a system-size check, Appendix A also presents dynamical comparison results for the $N = 12$ lattice system. Similar to the $N = 8$ case, the spatial charge distribution in the $N = 12$ system still shows boundary charge separation induced by the external electric field, indicating that this phenomenon is not a

numerical artifact specific to the small $N = 8$ system. Under the same external field strength, $P_{\text{vac}}(t)$ in the $N = 12$ system is more sensitive to the external field and deviates from its initial value more rapidly. This may be related to the denser low-energy spectrum and the larger number of excited states participating in the quench dynamics in the larger system.

V. SUMMARY AND OUTLOOK

In this work, we studied the static ground-state structure and nonequilibrium dynamics of the finite-lattice Schwinger model under a constant external

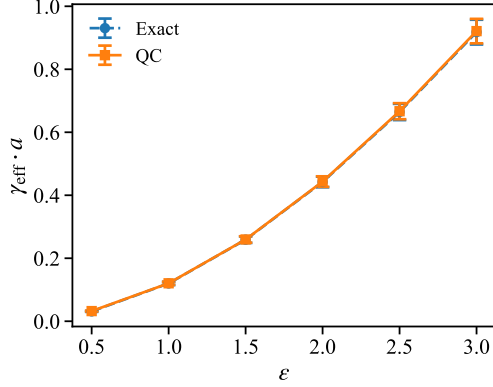


FIG. 7. Effective decay rate γ_{eff} obtained by fitting

electric field. We used VQE and VQD to calculate the low-energy states and employed a second-order Trotter-Suzuki decomposition to simulate the real-time evolution after an electric-field quench. The quantum-algorithm results were compared with exact diagonalization throughout the calculation.

The external-field scan shows step-like changes in the chiral condensate near the field strengths where the ground state and the first excited state become closest in energy. The corresponding characteristic field strength exhibits a clear finite-size dependence. After the electric-field quench, the total charge and total energy remain approximately conserved. The system exhibits boundary charge separation, quasiperiodic electric-field-energy redistribution, and a field-dependent decrease of the vacuum-state fidelity. The site-resolved charge dynamics and the $N = 12$ results further show that the main dynamical features persist for a larger lattice.

These results demonstrate that VQE-based state preparation combined with digital real-time evolution can reproduce the main static and dynamical responses of the finite-lattice Schwinger model. Future work may extend the present study to larger systems, improved variational ansatzes, noisy quantum hardware, and time-dependent external fields.

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Appendix A: Site-Resolved Charge Dynamics and the $N = 12$ Lattice

The real-time dynamics in the main text mainly focus on the $N = 8$ lattice and use the spatial-point charge $Q_i(t)$ to show the field-induced charge separation. In this appendix, we provide the corresponding site-resolved charge dynamics and the $N = 12$ lattice check. The line and marker conventions are the same as in the main text.

Figure 8 shows the time evolution of the staggered-site charge $q_n(t)$ for the $N = 8$ lattice. Compared with the spatial-point charge $Q_i(t)$ in the main text, the site-resolved charge contains more detailed staggered-component oscillations. The dominant response still appears near the open boundaries.

Figure 9 shows the corresponding site-resolved charge dynamics for the $N = 12$ lattice. The larger system contains more local oscillatory components, but the boundary-dominated response remains similar to the $N = 8$ case.

For comparison with the spatial-point observable used in the main text, Figure 10 shows $Q_i(t)$ for the $N = 12$ lattice. The result shows the same qualitative boundary charge separation as in the $N = 8$ lattice.

We also compare the vacuum-state fidelity for the $N = 12$ lattice. As shown in Figure 11, $P_{\text{vac}}(t)$ deviates from its initial value more rapidly than in the $N = 8$ case under the same external-field strength. This behavior may be related to the denser low-energy spectrum and the larger number of excited states involved in the quench dynamics.

Overall, the $N = 8$ and $N = 12$ results show that the boundary charge separation and the deviation from the initial vacuum state persist as the system size is increased.

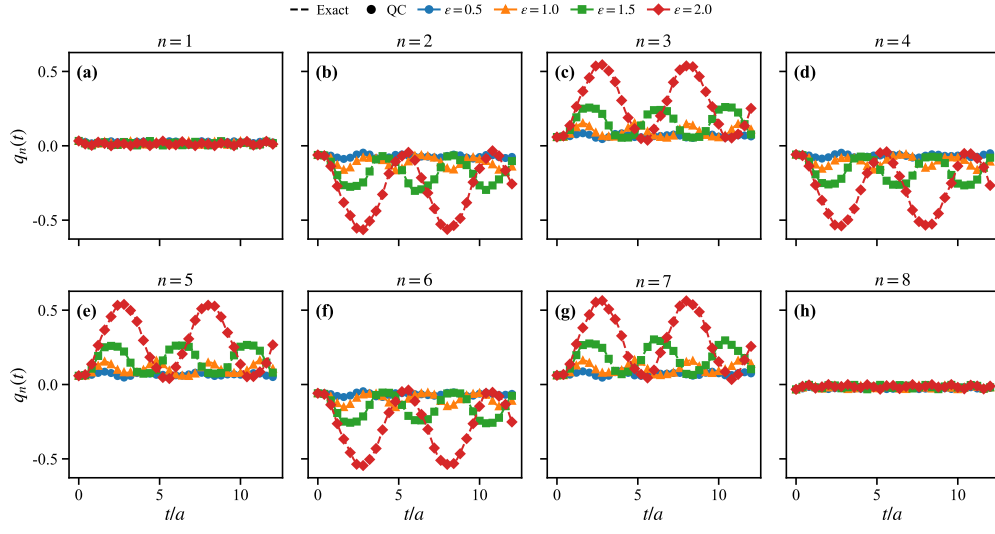


FIG. 8. Time evolution of the site-resolved charge $q_n(t)$ in the $N = 8$ lattice.

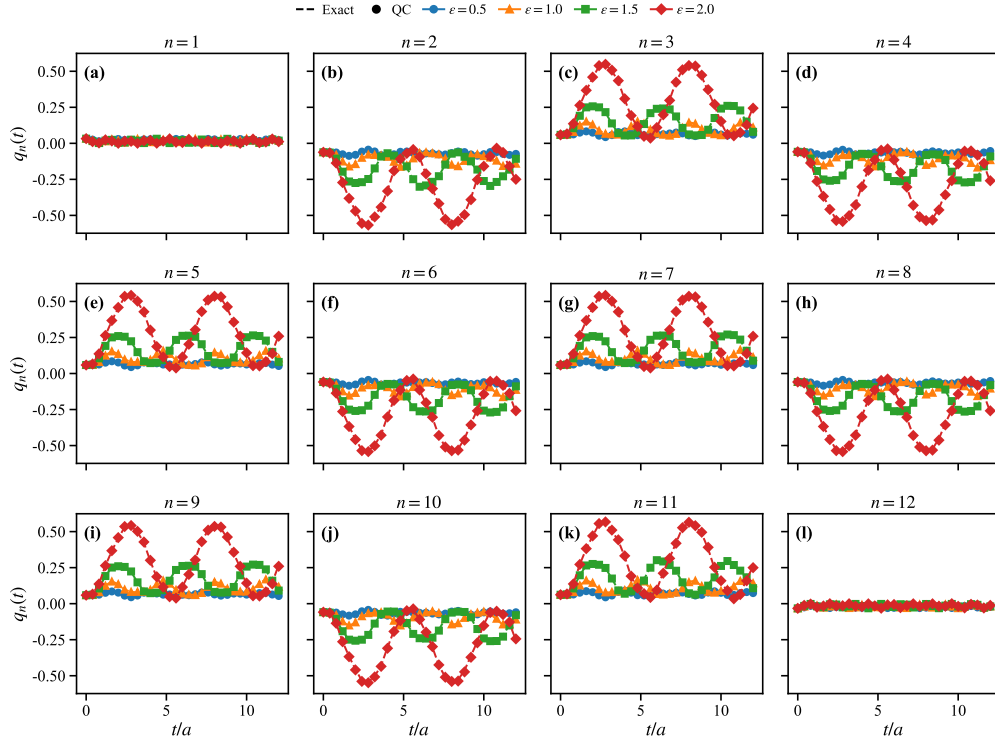


FIG. 9. Time evolution of the site-resolved charge $q_n(t)$ in the $N = 12$ lattice.

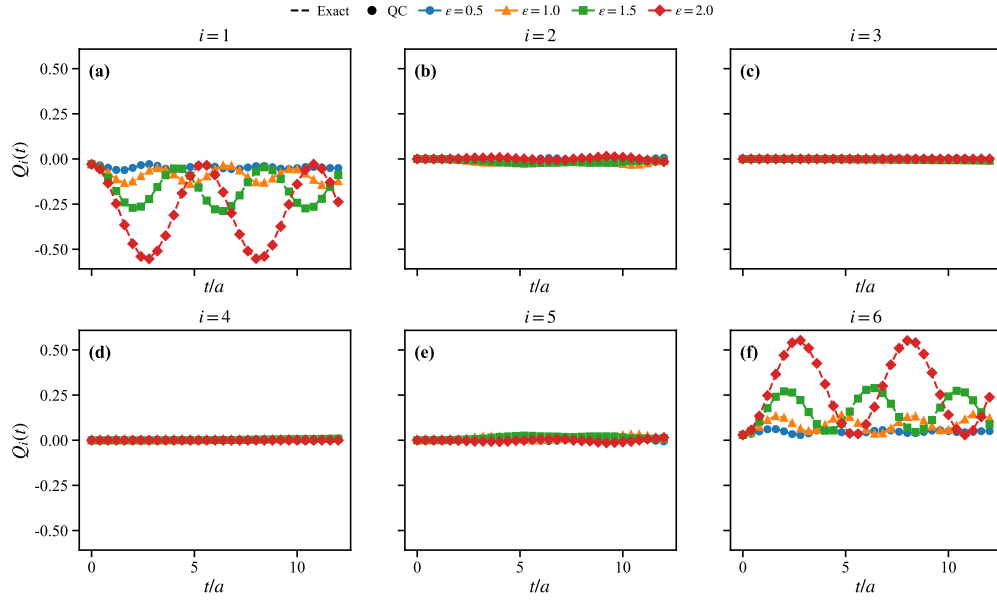


FIG. 10. Time evolution of the spatial-point charge $Q_i(t)$ in the $N = 12$ lattice.

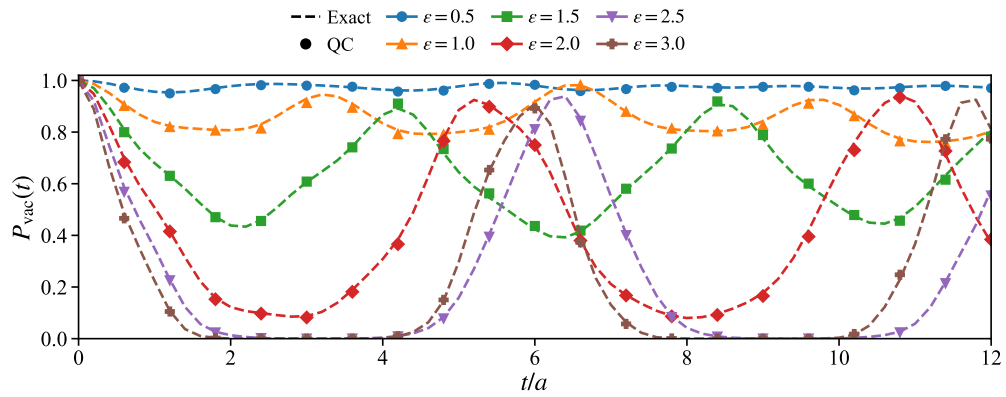


FIG. 11. Time evolution of the vacuum-state fidelity $P_{\text{vac}}(t)$ in the $N = 12$ lattice.

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